

and the division of β_t in Eqn 28 normalizes the gradient norm at each position t as we will later see in Eqn 31 and also clarified in Section 4.1 (and Appendix D.2).

Note that $\beta_t \cdot D_{\text{KL}}\left(\pi^*(\cdot|\mathbf{x}, \mathbf{y}_{<t}) \parallel \pi_{\text{aligned}}(\cdot|\mathbf{x}, \mathbf{y}_{<t})\right) \geq 0$ is a non-negative constant, we have:

$$\mathbb{P}_\theta\left(T([\mathbf{x}, \mathbf{y}_{<t}], y_t)\right) \geq \sigma\left(\beta_t \cdot \log \frac{\pi_\theta(y_t|\mathbf{x}, \mathbf{y}_{<t})}{\pi_{\text{aligned}}(y_t|\mathbf{x}, \mathbf{y}_{<t})}\right) \quad (30)$$

So, the objective in Eqn 28 can be replaced with a lower bound surrogate objective L_θ :

$$\mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \left\{ \sum_{t \geq 1} \frac{1}{\beta_t} \log \mathbb{P}_\theta\left(T([\mathbf{x}, \mathbf{y}_{<t}], y_t)\right) \right\} \geq L_\theta \quad (31)$$

where

$$L_\theta := \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \left\{ \sum_{t \geq 1} \frac{1}{\beta_t} \cdot \log \sigma\left(\beta_t \cdot \log \frac{\pi_\theta(y_t|\mathbf{x}, \mathbf{y}_{<t})}{\pi_{\text{aligned}}(y_t|\mathbf{x}, \mathbf{y}_{<t})}\right) \right\}.$$

Eqn 3 is then equivalent to $\min_\theta -L_\theta \iff \max_\theta L_\theta$. So, optimizing the objective Eqn 3 is essentially to maximize the lower-bound of the reinforcement learning objective in Eqn 28.